Arjun P

Embedded Software Engineer

arjunp1820@gmail.com | 6382123232 | Coimbatore, India | linkedin.com/in/arjunpalpandian

CAREER OBJECTIVE

Embedded software engineer with hands-on experience designing microcontroller-based firmware, sensor integration, and real-time control logic across multiple IoT and automation projects. Proficient in C/C++ for embedded targets, with practical exposure to sensor interfacing, wireless communication protocols, and computer vision integration. Seeking to apply strong debugging, systems-level thinking, and rapid prototyping skills to build reliable embedded software for smart mobility and connected-device platforms.

TECHNICAL SKILLS

Embedded Programming: C, C++, Arduino, microcontroller firmware development, register-level peripheral configuration

Embedded Platforms: Arduino, microcontroller-based boards, sensor-driven embedded systems

Communication Protocols: LoRa (long-range wireless), GSM, WiFi (IMU data streaming), I2C/serial sensor interfacing

Sensors & Actuators: Ultrasonic sensors, IMU (accelerometer/gyroscope), solenoid actuation, computer vision-based positional sensing

Systems & Tools: Linux, VS Code, Tinkercad, Wokwi (circuit simulation), Git-based workflows

Software Practices: Real-time data acquisition, state-machine/process-control logic, interlock and fault-detection logic, cloud/dashboard data logging

PROJECTS

Smart Assembly Monitoring System

- Designed a configurable, software-driven process-control algorithm supporting operator-defined reference initiation, dynamic step validation, visual guidance indicators, and automatic interlock simulation triggered on sequence deviation.
- Built a hybrid sensing and firmware architecture fusing computer vision-based positional mapping with smartphone IMU (accelerometer/gyroscope) data streamed over WiFi, enabling real-time tool orientation tracking and spatial zone identification.
- Implemented the embedded logic layer responsible for real-time state tracking, sensor fusion, and deviation handling between hardware inputs and the control algorithm.

Water Distribution System (LoRa-Enabled)

- Developed firmware for a LoRa-enabled embedded system performing real-time water-level acquisition and long-range, low-power data transmission across municipal zones.
- Programmed microcontroller-based embedded architecture integrating ultrasonic sensors and LoRa modules for continuous sensing and remote telemetry.
- Implemented cloud/remote dashboard connectivity in firmware for data logging, threshold-based alerts, and performance analysis, ensuring reliable low-power communication.

Braille Display for Inclusive Education

- Developed embedded control software for a tactile Braille communication system, driving solenoid-based pins to convert digital text into real-time Braille patterns.
- Implemented precise firmware-level actuation control to ensure accurate pin timing and movement, enabling seamless character-to-pattern translation.
- Built a microcontroller-driven character-mapping system in firmware to store Braille patterns and manage fast, reliable updates for continuous reading.

Smart Garbage Monitoring System

- Developed embedded firmware on Arduino integrating GSM and ultrasonic sensing with automated lid-actuation control logic.
- Implemented real-time fill-level detection algorithms and hands-free lid control through sensor-driven firmware logic.

- Programmed SMS alert triggers in firmware once fill thresholds were reached, improving collection efficiency and reducing manual checks.

INTERNSHIP EXPERIENCE

Unartech Solutions *05/2022 | Tirupur*

- Contributed to the development and integration of computer vision models for image detection, supporting software-driven workflow improvements and accuracy gains.
- Collaborated on debugging and refining detection logic to improve reliability of automated image-processing workflows.

Bharat Automation *12/2024 | Coimbatore*

- Updated and maintained project tracking systems to reflect real-time status and scope changes for an embedded municipal water management system.
- Supported coordination between hardware and software teams to track firmware and system-level changes through project lifecycle.

Alstom Transport India Limited *01/2025 | Coimbatore*

- Performed quality inspections on electrical components, gaining exposure to system-level compliance standards relevant to embedded hardware reliability.

Fuji Electric *09/2025 | Chennai*

- Performed electrical tests on transformers to verify compliance with industry standards, building a strong foundation in hardware-software interaction and signal verification.

EDUCATION

B.E. - Electrical and Electronics Engineering, Karpagam Institute of Technology *2022 - 2026 | Coimbatore*
CGPA: 7.33

Higher Secondary Education, Kalaimagal Higher Secondary School *2021 - 2022 | Sivakasi*
Percentage: 84%

CERTIFICATIONS

- Actively participated in an IoT and Embedded Systems workshop at Hindustan College, gaining practical exposure to real-time embedded development.
- Successfully completed the NPTEL course on Internet of Things with 62% score, achieving ELITE grade certification.
- Successfully completed Arduino Training through IIT Spoken Tutorial with 75% score.
- Completed Embedded System Design training at Vector India, focusing on project-oriented learning and real-time application development.
- Participated in the Smart Garbage Monitoring System project showcase at Karpagam Institute of Technology, presenting an IoT-based automation solution.

SOFT SKILLS

- Continuous Learning Mindset
- Structured Problem-Solving & Debugging
- Operational Adaptability
- Rapid Skill Acquisition